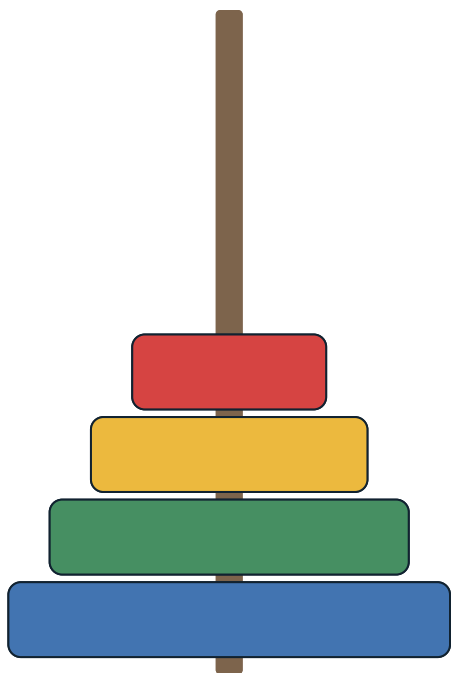


TORRE DE HANÓI

Minha solução: quatro discos, três hastes e 15 movimentos



H1



H2



H3

Sequência visual dos estados, do início ao objetivo em H2.

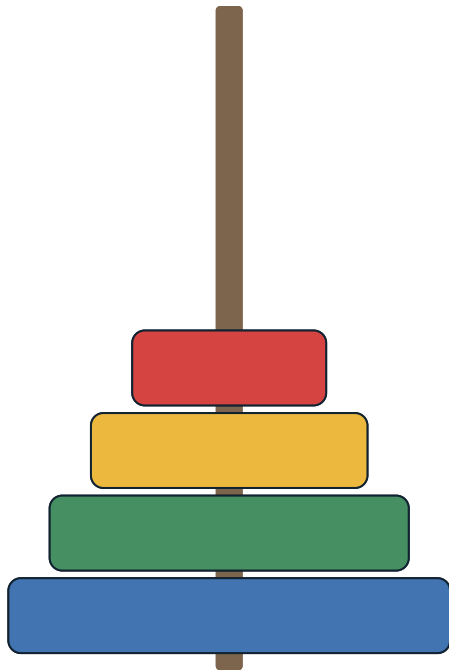
ENUNCIADO E REGRAS

Objetivo: transferir os quatro discos de H1 para H2.

1. Mover apenas um disco por vez.
2. Nunca colocar um disco maior sobre um menor.
3. Utilizar H3 como haste auxiliar sempre que necessário.
4. Registrar cada configuração após o movimento.

Cores e tamanhos: vermelho (menor), amarelo, verde e azul (maior).

Quatro discos empilhados na haste H1.



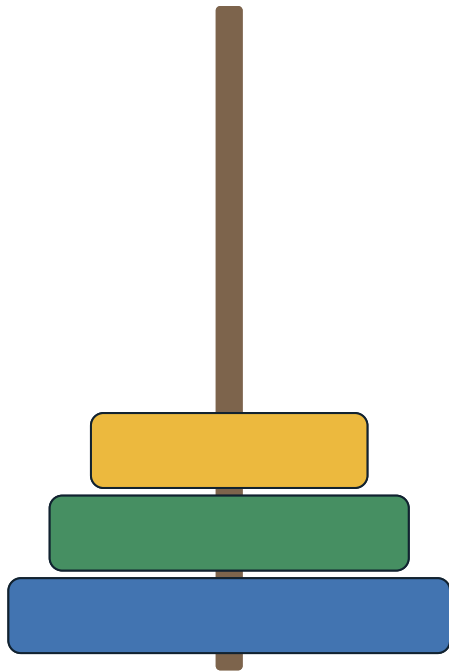
H1

H2

H3

Estado 0/15 H1: [4, 3, 2, 1] H2: [] H3: []

Mover o disco 1 (vermelho) de H1 para H3.



H1



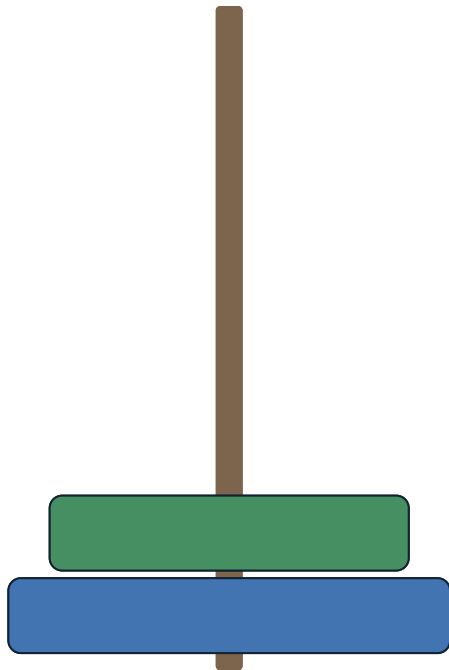
H2



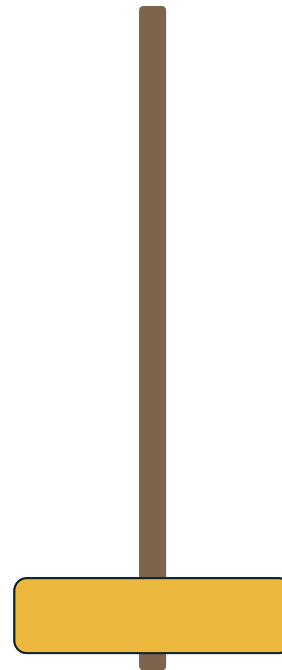
H3

Estado 1/15 H1: [4, 3, 2] H2: [] H3: [1]

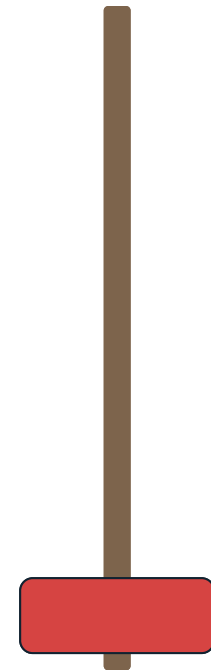
Mover o disco 2 (amarelo) de H1 para H2.



H1



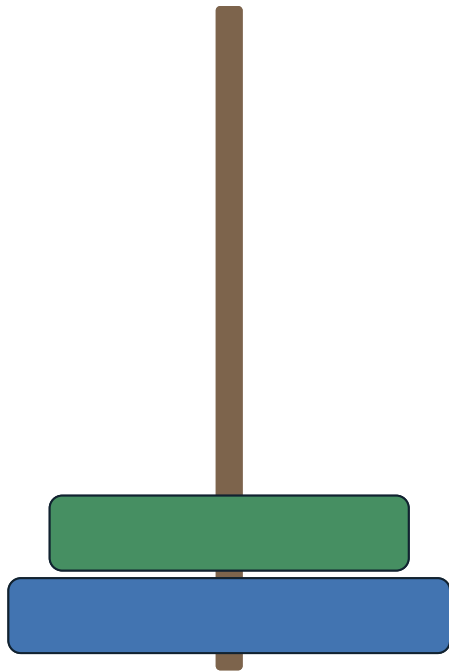
H2



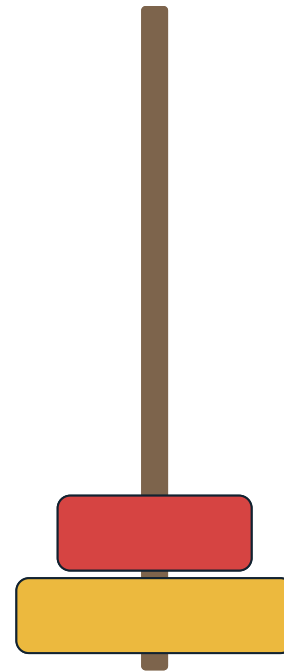
H3

Estado 2/15 H1: [4, 3] H2: [2] H3: [1]

Mover o disco 1 (vermelho) de H3 para H2.



H1



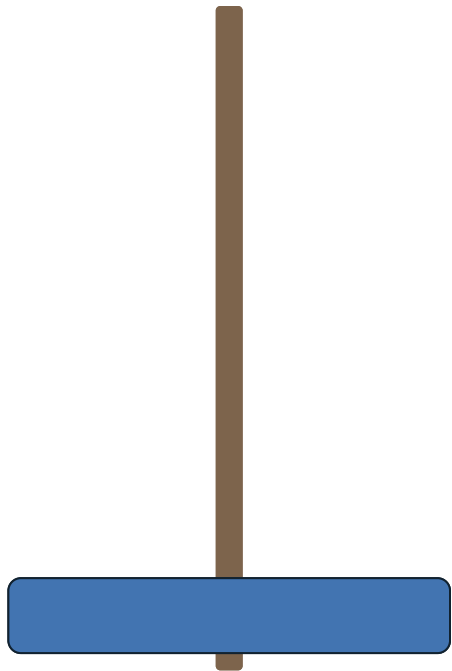
H2



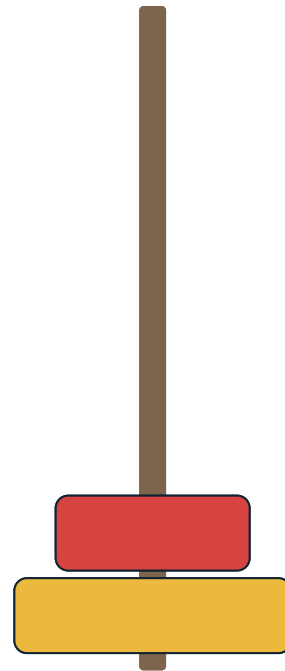
H3

Estado 3/15 H1: [4, 3] H2: [2, 1] H3: []

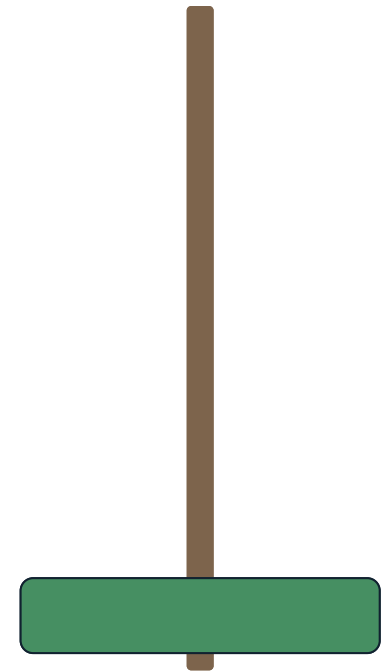
Mover o disco 3 (verde) de H1 para H3.



H1



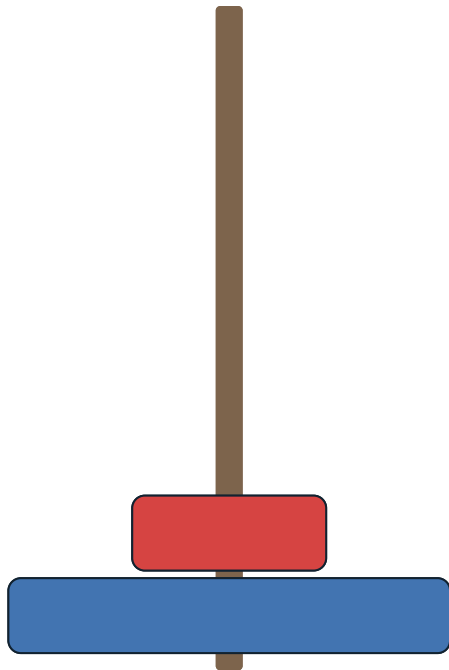
H2



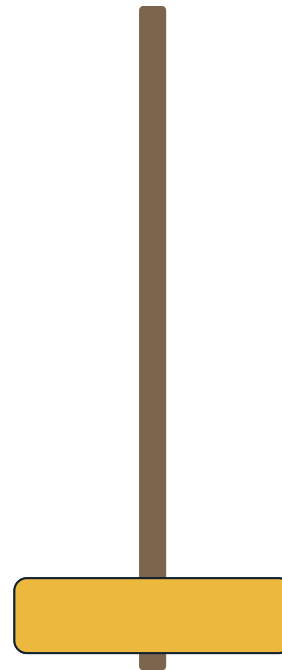
H3

Estado 4/15 H1: [4] H2: [2, 1] H3: [3]

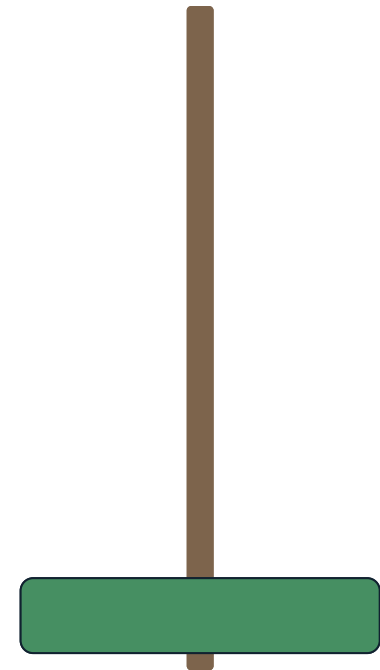
Mover o disco 1 (vermelho) de H2 para H1.



H1



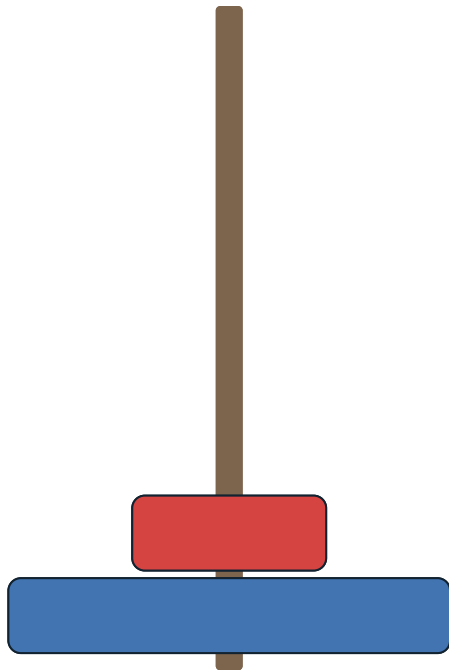
H2



H3

Estado 5/15 H1: [4, 1] H2: [2] H3: [3]

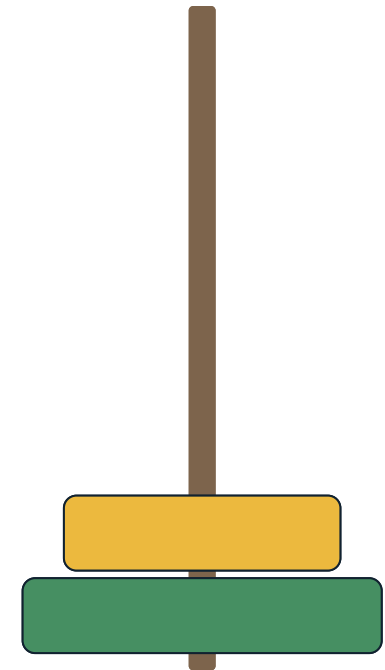
Mover o disco 2 (amarelo) de H2 para H3.



H1



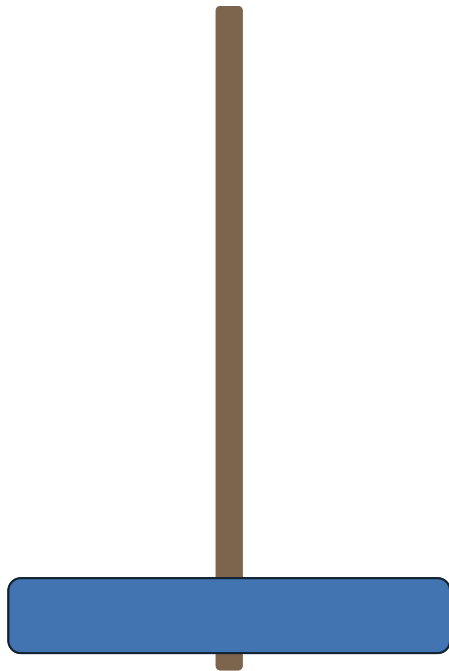
H2



H3

Estado 6/15 H1: [4, 1] H2: [] H3: [3, 2]

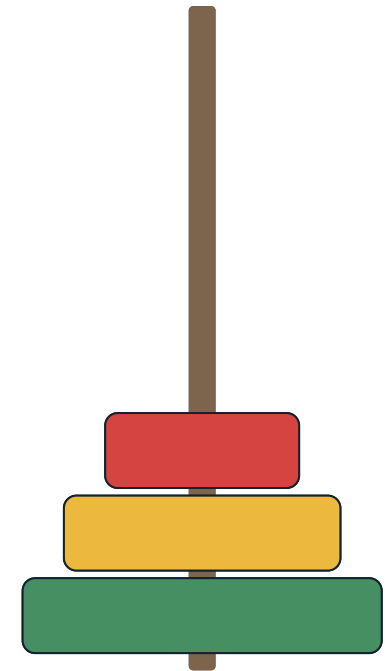
Mover o disco 1 (vermelho) de H1 para H3.



H1



H2



H3

Estado 7/15 H1: [4] H2: [] H3: [3, 2, 1]

Mover o disco 4 (azul) de H1 para H2.



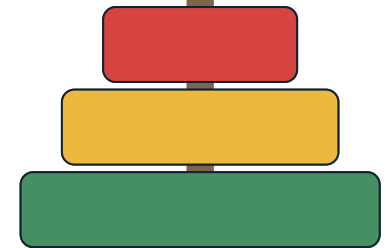
H1



H2



H3

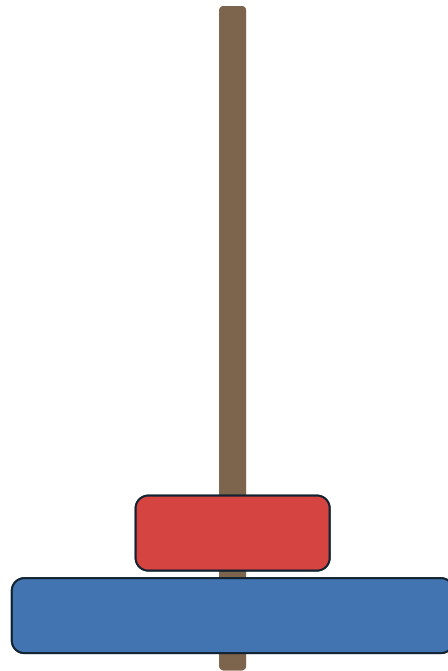


Estado 8/15 H1: [] H2: [4] H3: [3, 2, 1]

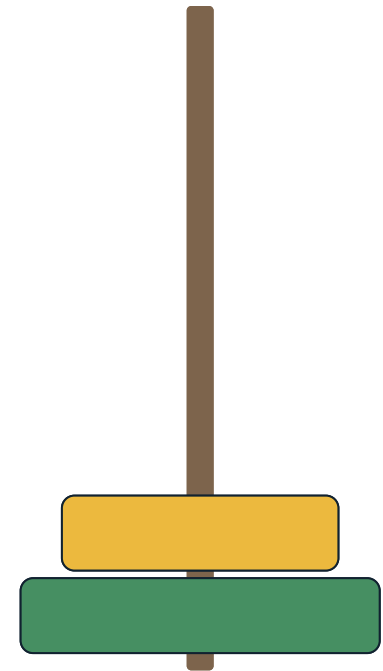
Mover o disco 1 (vermelho) de H3 para H2.



H1



H2



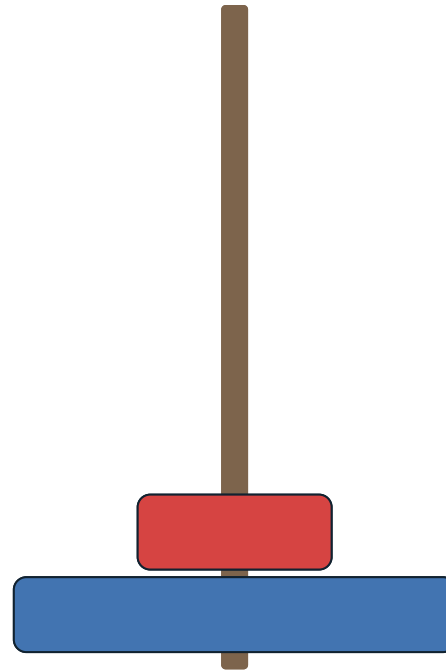
H3

Estado 9/15 H1: [] H2: [4, 1] H3: [3, 2]

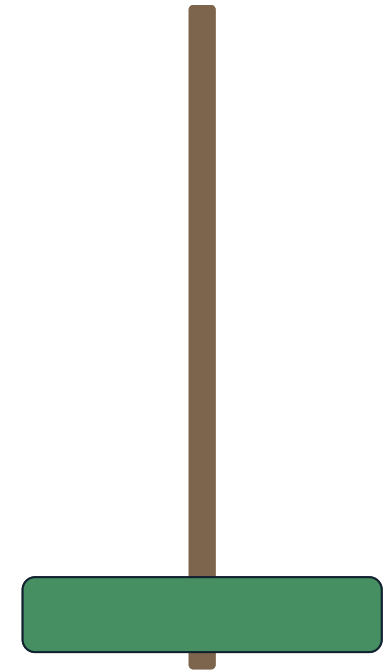
Mover o disco 2 (amarelo) de H3 para H1.



H1



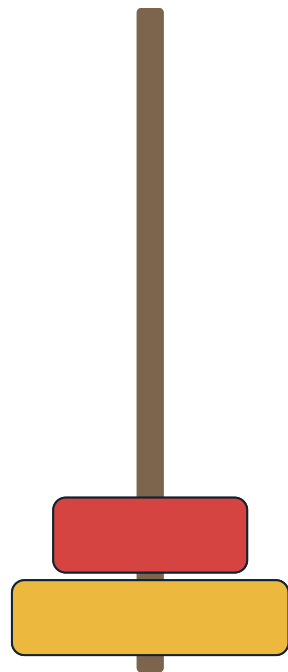
H2



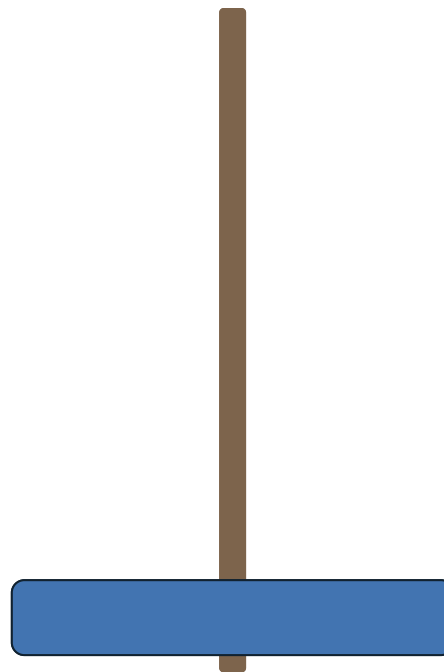
H3

Estado 10/15 H1: [2] H2: [4, 1] H3: [3]

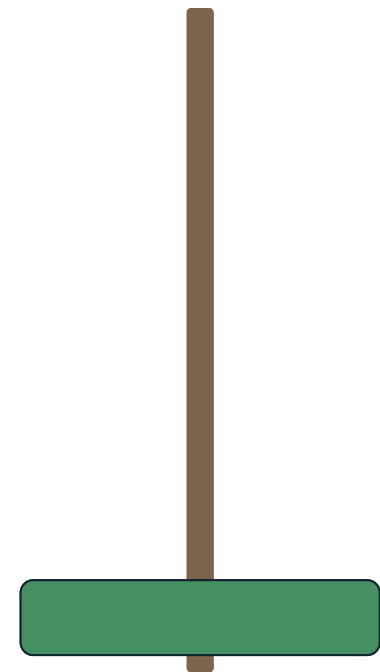
Mover o disco 1 (vermelho) de H2 para H1.



H1



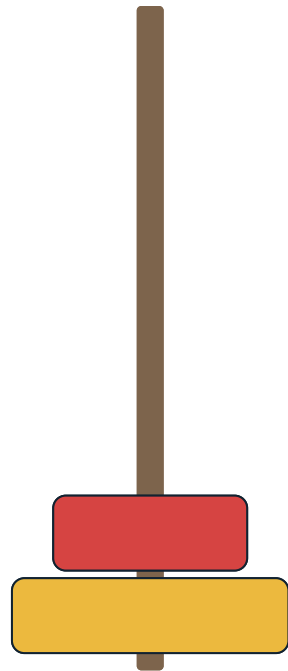
H2



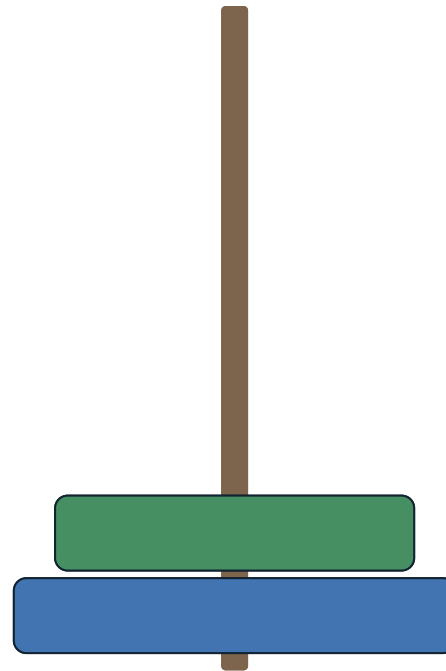
H3

Estado 11/15 H1: [2, 1] H2: [4] H3: [3]

Mover o disco 3 (verde) de H3 para H2.



H1



H2



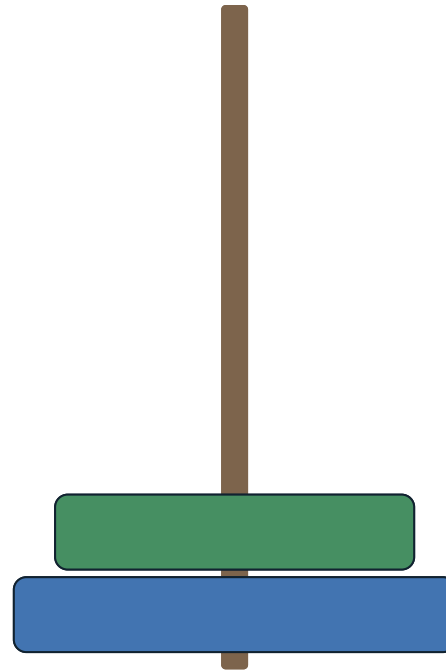
H3

Estado 12/15 H1: [2, 1] H2: [4, 3] H3: []

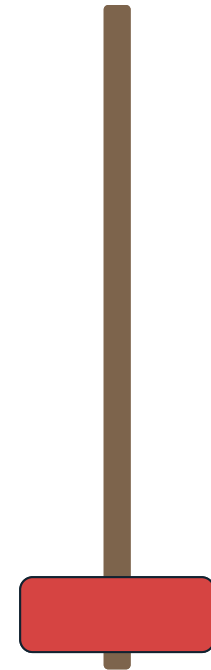
Mover o disco 1 (vermelho) de H1 para H3.



H1



H2



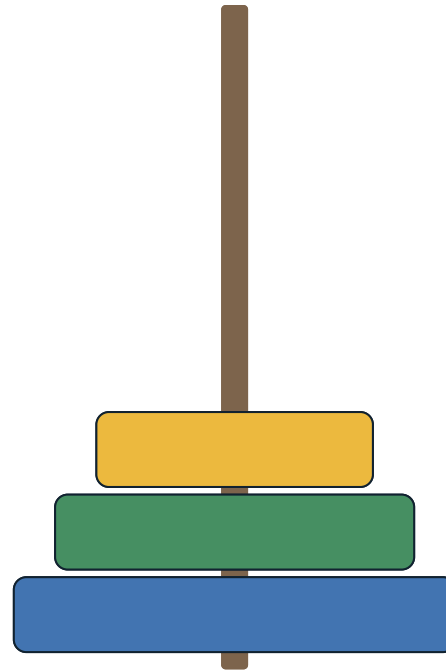
H3

Estado 13/15 H1: [2] H2: [4, 3] H3: [1]

Mover o disco 2 (amarelo) de H1 para H2.



H1



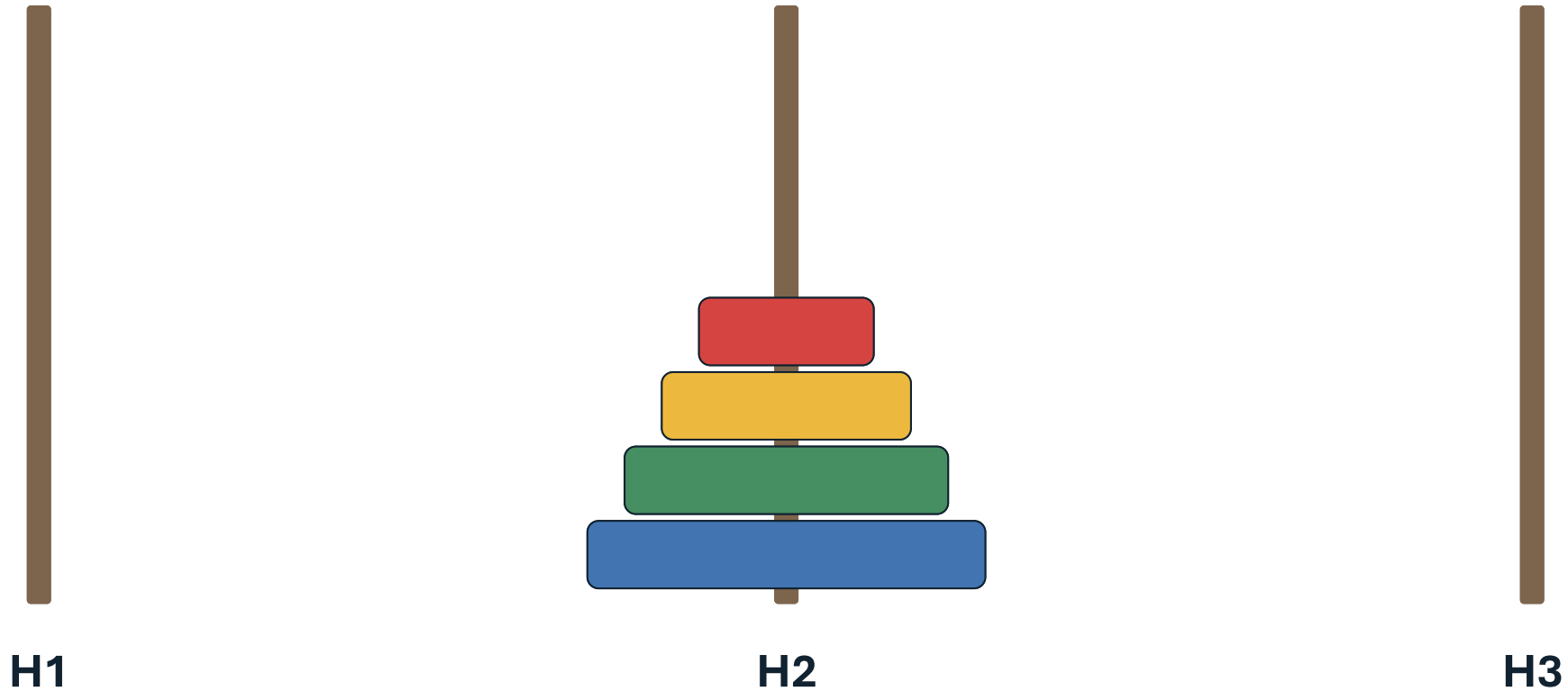
H2



H3

Estado 14/15 H1: [] H2: [4, 3, 2] H3: [1]

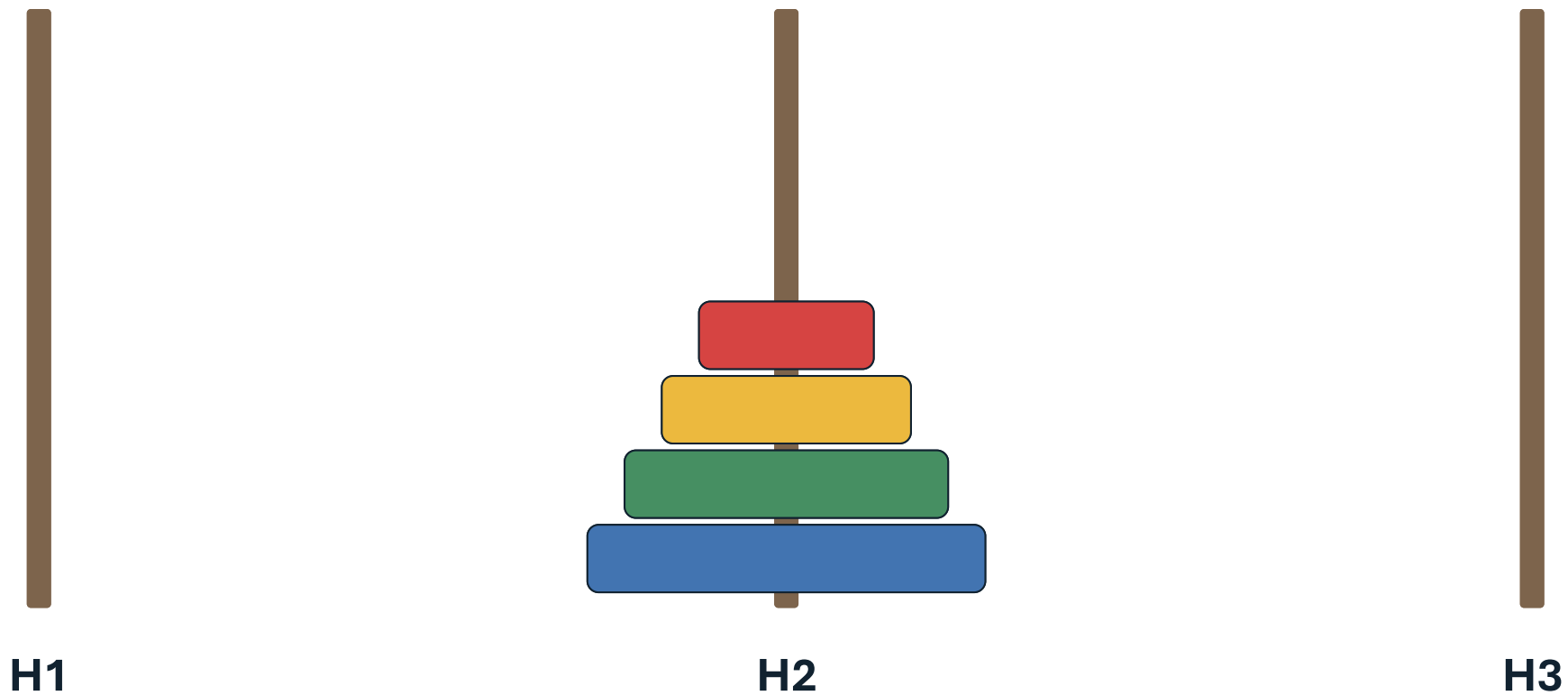
Mover o disco 1 (vermelho) de H3 para H2.



Estado 15/15 H1: [] H2: [4, 3, 2, 1] H3: []

RESULTADO E RACIOCÍNIO

Os quatro discos chegaram à haste H2 em 15 movimentos.



$$M(4) = 2 \times M(3) + 1 = 2 \times 7 + 1 = 15$$